

Rectangular Harmonic Congruences in Inert Imaginary Quadratic Orders

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Abstract

Let $\omega^2 - T\omega + N = 0$ have negative discriminant $\Delta = T^2 - 4N$, and let a prime $p > 5$ be unramified and inert in $\mathbb{Z}[\omega]$. We prove that the inverse first and second moments over the endpoint-inclusive p^k -by- p^k rectangle of p -adic units satisfy

$$H_{1,\omega}(k) \in p^{2k}\mathcal{O}_{\omega,p}, \quad H_{2,\omega}(k) \in p^k\mathcal{O}_{\omega,p}$$

for every $k \geq 1$. At level one, pairing a positive-coordinate representative with its negative reduces the missing factor of p to three vanishing inverse-square sums over finite fields. An exact coordinate-block decomposition then propagates both congruences simultaneously. This proves the general imaginary-quadratic-order assertion formulated as Conjecture 6.3 by Calderon; exact quotient-ring computations are reported only as corroboration.

1 Statement and context

Fix $T, N \in \mathbb{Z}$ with

$$\omega^2 - T\omega + N = 0, \quad \Delta = T^2 - 4N < 0,$$

and put $\mathcal{O}_\omega = \mathbb{Z}[\omega]$. Let $p > 5$ be a prime such that $p \nmid \Delta$ and

$$\left(\frac{\Delta}{p}\right) = -1.$$

Thus $X^2 - TX + N$ is irreducible modulo p . We write $\mathcal{O}_{\omega,p}$ for the local ring at p . For $k \geq 1$, define

$$U_{\omega,k} = \{x + y\omega : 1 \leq x, y \leq p^k, p \nmid \gcd(x, y)\}$$

and, for $r \geq 1$,

$$H_{r,\omega}(k) = \sum_{z \in U_{\omega,k}} z^{-r} \in \mathcal{O}_{\omega,p}.$$

Every inverse exists: reduction modulo p identifies $\mathcal{O}_\omega/p\mathcal{O}_\omega$ with $\mathbb{F}_{p^2}$, and the unit condition says exactly that the two residue coordinates are not both zero.

Theorem 1. *Under the preceding hypotheses, for every integer $k \geq 1$,*

$$H_{1,\omega}(k) \in p^{2k}\mathcal{O}_{\omega,p}, \quad H_{2,\omega}(k) \in p^k\mathcal{O}_{\omega,p}.$$

Calderon [1] proved these bounds for the Gaussian order and inert Gaussian primes, and stated the order-uniform assertion above as Conjecture 6.3. The point not supplied by the residue-field identity $\sum_{v \neq 0} v^{-1} = 0$ alone is the second factor of p in $H_{1,\omega}(1)$. Lemma 2 obtains it while retaining the positive-coordinate representative convention. Lemma 3 then isolates the low power sums needed for a uniform lift in k . The comparison here is with that cited source; it is not a general priority claim. For broad adjacent context, finite-prime restrictions on pairs of lattice coordinates also occur in a different arithmetic lattice-graph problem [2]; that work supplies no input to the present theorem.

2 The level-one pairing

Put

$$\mathbb{F} = \mathcal{O}_\omega / p\mathcal{O}_\omega \cong \mathbb{F}_{p^2}.$$

For $a \in \mathbb{F}_p$, let $\tilde{a} \in \{1, \dots, p\}$ denote its positive representative, with $\tilde{0} = p$. If $v = a + b\omega \in \mathbb{F}^\times$, set

$$s(v) = \tilde{a} + \tilde{b}\omega.$$

The elements $s(v)$, as v ranges over $\mathbb{F}^\times$, are exactly the members of $U_{\omega,1}$.

Lemma 2. *One has*

$$H_{1,\omega}(1) \in p^2\mathcal{O}_{\omega,p}, \quad H_{2,\omega}(1) \in p\mathcal{O}_{\omega,p}.$$

Proof. Since p is odd, $v \mapsto -v$ has no fixed point on $\mathbb{F}^\times$. Coordinatewise,

$$\tilde{a} + \widetilde{-a} = p(1 + \mathbf{1}_{\{a=0\}}).$$

Consequently

$$s(v) + s(-v) = pc(v), \quad c(v) = 1 + \omega + \mathbf{1}_{\{a=0\}} + \omega\mathbf{1}_{\{b=0\}}.$$

Writing $s(-v) = pc(v) - s(v)$ and expanding the inverse only modulo p^2 gives the exact pair congruence

$$s(v)^{-1} + s(-v)^{-1} \equiv -pc(v)s(v)^{-2} \pmod{p^2\mathcal{O}_{\omega,p}}.$$

The functions $c(v)$ and v^{-2} are unchanged by replacing v with $-v$. Hence, because 2 is a unit,

$$H_{1,\omega}(1) \equiv -\frac{p}{2} \sum_{v \in \mathbb{F}^\times} c(v)v^{-2} \pmod{p^2\mathcal{O}_{\omega,p}}, \tag{1}$$

where the coefficient after p is evaluated in $\mathbb{F}$.

Separate the two coordinate axes in that coefficient. Suppressing residue bars on ω , we obtain

$$\sum_{v \in \mathbb{F}^\times} c(v)v^{-2} = (1 + \omega) \sum_{v \in \mathbb{F}^\times} v^{-2} + \omega^{-2} \sum_{b \in \mathbb{F}_p^\times} b^{-2} + \omega \sum_{a \in \mathbb{F}_p^\times} a^{-2}. \tag{2}$$

Here $\omega \neq 0$ in $\mathbb{F}$, since the defining quadratic is irreducible. Inversion permutes each multiplicative group. The sum of squares in $\mathbb{F}_q^\times$ is zero whenever $q - 1 \nmid 2$; apply this for $q = p^2$ and $q = p$. Since $p > 5$, every sum on the right of (2) therefore vanishes. Equation (1) proves the first assertion.

For the second assertion, direct reduction modulo p gives

$$H_{2,\omega}(1) \equiv \sum_{v \in \mathbb{F}^\times} v^{-2} = 0 \pmod{p\mathcal{O}_{\omega,p}}.$$

□

3 Coordinate blocks and their power sums

For $k \geq 2$, put $Q = p^{k-1}$ and

$$D = \{i + j\omega : 0 \leq i, j < p\}.$$

The endpoint convention in $U_{\omega,k}$ is important here. Every $x \in \{1, \dots, p^k\}$ has a unique representation $x = x_0 + Qi$, where $1 \leq x_0 \leq Q$ and $0 \leq i < p$; this also represents $x = p^k$ uniquely, with $x_0 = Q$ and $i = p - 1$.

Lemma 3. *For every $k \geq 2$ there is an exact disjoint decomposition*

$$U_{\omega,k} = \{u + Qw : u \in U_{\omega,k-1}, w \in D\}. \quad (3)$$

If

$$S_m = \sum_{w \in D} w^m,$$

then

$$S_0 = p^2, \quad S_m \in p^2 \mathcal{O}_{\omega} \quad (1 \leq m \leq 3). \quad (4)$$

Proof. Apply the preceding coordinate representation to both coordinates. Since $p \mid Q$, the two original coordinates are simultaneously divisible by p if and only if the two lower-block coordinates are. This proves (3), including disjointness.

The identity $S_0 = p^2$ is immediate. For $1 \leq m \leq 3$, the binomial theorem gives

$$S_m = \sum_{h=0}^m \binom{m}{h} \omega^h \left(\sum_{i=0}^{p-1} i^{m-h} \right) \left(\sum_{j=0}^{p-1} j^h \right). \quad (5)$$

For every $0 \leq d \leq 3$, the integer $\sum_{t=0}^{p-1} t^d$ is divisible by p : this is clear for $d = 0$, and for $1 \leq d \leq 3 < p - 1$ its reduction is the corresponding vanishing power sum in $\mathbb{F}_p$. Thus both parenthesized factors in every summand of (5) are divisible by p , proving (4). $\square$

4 Simultaneous lifting

For bookkeeping, define

$$A_m(\ell) = \sum_{u \in U_{\omega,\ell}} u^{-m} \quad (m \geq 1).$$

Thus $A_1 = H_{1,\omega}$ and $A_2 = H_{2,\omega}$, while every $A_m(\ell)$ is integral at p . For $u \in U_{\omega,k-1}$ and $w \in D$, the element Qwu^{-1} belongs to $p\mathcal{O}_{\omega,p}$. We may therefore use a finite geometric expansion of $(1 + Qwu^{-1})^{-r}$; after truncation in degree d , its remainder is $(Qwu^{-1})^{d+1}$ times an element of $\mathcal{O}_{\omega,p}$.

We now prove Theorem 1 by simultaneous induction. The base case is Lemma 2. Assume the asserted bounds at level $k - 1$. Using (3), expanding the second inverse moment through degree one, and then summing first over w yields

$$H_{2,\omega}(k) = p^2 H_{2,\omega}(k-1) - 2QS_1 A_3(k-1) + R_2, \quad R_2 \in Q^2 \mathcal{O}_{\omega,p}. \quad (6)$$

The three terms in (6) are respectively in

$$p^{k+1} \mathcal{O}_{\omega,p}, \quad p^{k+1} \mathcal{O}_{\omega,p}, \quad p^{2(k-1)} \mathcal{O}_{\omega,p} \subseteq p^k \mathcal{O}_{\omega,p}.$$

This proves the second-moment bound at level k .

For the first moment and $k \geq 3$, expand through degree two:

$$\begin{aligned} H_{1,\omega}(k) &= p^2 H_{1,\omega}(k-1) - Q S_1 H_{2,\omega}(k-1) \\ &\quad + Q^2 S_2 A_3(k-1) + R_1, \quad R_1 \in Q^3 \mathcal{O}_{\omega,p}. \end{aligned} \quad (7)$$

Using the induction hypothesis and (4), the four displayed terms have p -adic valuation at least

$$2 + 2(k-1), \quad (k-1) + 2 + (k-1), \quad 2(k-1) + 2, \quad 3(k-1),$$

respectively. All four lower bounds are at least $2k$.

The exceptional lift from $k = 1$ to $k = 2$ requires one additional term. Expanding through degree three gives

$$\begin{aligned} H_{1,\omega}(2) &= p^2 H_{1,\omega}(1) - p S_1 H_{2,\omega}(1) + p^2 S_2 A_3(1) \\ &\quad - p^3 S_3 A_4(1) + R'_1, \quad R'_1 \in p^4 \mathcal{O}_{\omega,p}. \end{aligned} \quad (8)$$

The first three terms and the remainder lie in $p^4 \mathcal{O}_{\omega,p}$; by (4), the degree-three term lies in $p^5 \mathcal{O}_{\omega,p}$. Equations (7) and (8) prove $H_{1,\omega}(k) \in p^{2k} \mathcal{O}_{\omega,p}$ for every k , completing the induction.

The proof is uniform in the chosen integral quadratic relation. Any reduction of powers of ω uses only $\omega^2 = T\omega - N$ with integral coefficients, so it cannot diminish the powers of p established above.

5 Scope, computation, and assistance

The theorem concerns precisely the endpoint-inclusive positive-coordinate rectangle, unramified inert primes $p > 5$, and the first two inverse moments. No assertion is made here for split or ramified primes, primes at most five, other representative regions, higher inverse moments, or stronger exponents.

Two deterministic quotient-ring scans were carried out with integer arithmetic modulo p^{2k} . Inversion used conjugation and the quadratic norm, and every computed inverse was multiplied back by its input. The first scan covered 234 admissible (T, N, Δ, p) tuples, 365 level rows, and 2,075,664 summands, using negative discriminants through -40 , $|T| \leq 4$, inert primes through 19, all cases at $k = 1$, and $k = 2$ when $p \leq 13$. An expanded held-out scan covered 1,058 tuples, 1,644 rows, and 19,694,856 summands, with negative discriminants through -80 , $|T| \leq 7$, primes through 29, and $k = 2$ when $p \leq 17$. No failure occurred. For the source example $(T, N, p, k) = (1, 2, 13, 1)$, the coordinate residues modulo 169 were

$$H_{1,\omega}(1) = (0, 0), \quad H_{2,\omega}(1) = (156, 26).$$

These finite calculations corroborate the statement but are not used in the proof.

AI systems assisted with proof search, algebraic and valuation checks, generation of the exact scanning code, and manuscript drafting. The computational checks used exact integer arithmetic rather than floating point. Neither AI output nor finite computation is invoked as a premise of Theorem 1; the complete deductive argument is given above.

References

- [1] K. Calderon, *Ljunggren–Jacobsthal and Bailey-Type Congruences for Rectangular Gaussian Binomial Coefficients*, arXiv:2608.00347v1 (2026), <https://arxiv.org/abs/2608.00347>.
- [2] Z. Zeng, *Finite-prime obstructions for Erdős Problem 1212*, Zenodo preprint (2026), [doi:10.5281/zenodo.22293614](https://doi.org/10.5281/zenodo.22293614).